USER & PROJECT GUIDE

eSSL Attendance Management System

For eSSL Classic and Orcus biometric machines: what the app does, how to use it, what is finished, and what is still to do.



Version: 1.1 (adds copying biometrics of selected employees)

Date: 28 October 2026

Code: github.com/amit5042k/Essl-Biometric, branch claude/blissful-cerf-j3m9mw

Status: Built and tested with device simulators. Not yet tested on a real Classic or Orcus machine.

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1. Summary in one page

This app collects punches (fingerprint, face or card) from eSSL biometric machines and turns them into attendance: who came, when, who was late, who was absent, hours worked, and overtime. HR uses it from any web browser to see daily attendance and print the monthly muster roll.

2 machine types supported (Classic and Orcus)	**2** connection methods (TCP/IP and ADMS cloud)	**10** screens in the app
7 attendance status codes (P, A, HD, MP, WO, HO, L)	**18** automated tests, all passing	**0** real machines tested so far (see section 9)

Ready now

Collecting punches from both machine types, copying fingerprints/faces of selected employees between machines, employees, shifts, holidays, leaves, manual punches, daily report, monthly muster, CSV/Excel export, admin login, phone-friendly screens, GoDaddy hosting instructions.

Needed before go-live

Test with your real machines, pick the hosting plan, set a strong admin password, use HTTPS for the web screens, and schedule database backups. Full list in section 9.

Who does what

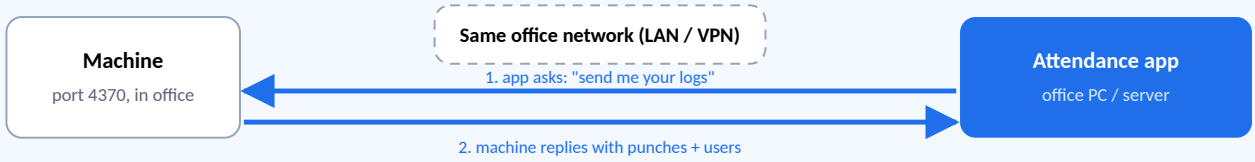
PERSON	WHAT THEY DO IN THE APP
HR / Admin	Adds employees, sets shifts and holidays, enters leaves, fixes missed punches, checks the daily report, exports the monthly muster for payroll.
IT person	Installs the app on a server or GoDaddy, sets the machines' network or cloud server settings, adds machines on the Devices page, takes backups.
Employees	Only punch on the machine. They have no login (self-service is a possible future feature).

2. How the app talks to the machines

eSSL machines can share their punches in two ways. The app supports both, and you can mix them (for example, one machine in the office on TCP/IP and one at the factory on ADMS).

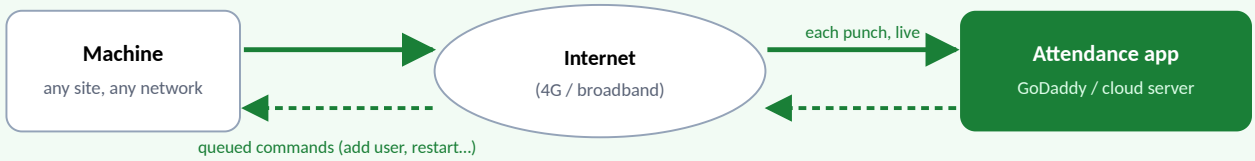
Method A – TCP/IP ("pull")

The app connects to the machine and downloads punches every few minutes.



Method B – ADMS / Cloud Server ("push")

The machine sends each punch to the app over the internet as soon as it happens.



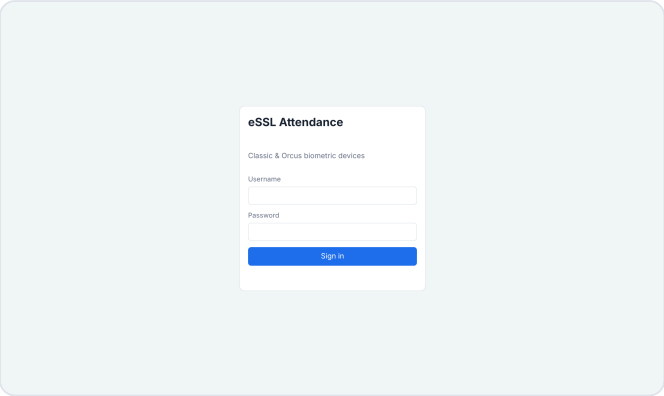
	A. TCP/IP (PULL)	B. ADMS / CLOUD (PUSH)
Who starts the connection	The app	The machine
Where the app can run	A PC or server in the same office as the machine	Anywhere on the internet, including GoDaddy
How fast punches appear	Every 5 minutes by default (adjustable), or straight away with "Sync now"	Within seconds
Machine menu to set	Comm. → Ethernet (fixed IP) and Comm Key	Comm. → Cloud Server Setting (ADMS)
Best for	A single office with a local PC that stays on	Several sites, remote branches, or cloud hosting

> **Important for GoDaddy hosting**
> A server on the internet cannot reach a machine inside your office network. If the app is hosted on GoDaddy, set the machines to **ADMS / Cloud Server** (method B). Check that your machines have this menu; Orcus and most recent Classic firmware do.

3. A tour of every screen

Open the app in any browser (Chrome, Edge, Firefox, or a phone). The menu on the left takes you to each screen. All screenshots below are from the real app, filled with sample data.

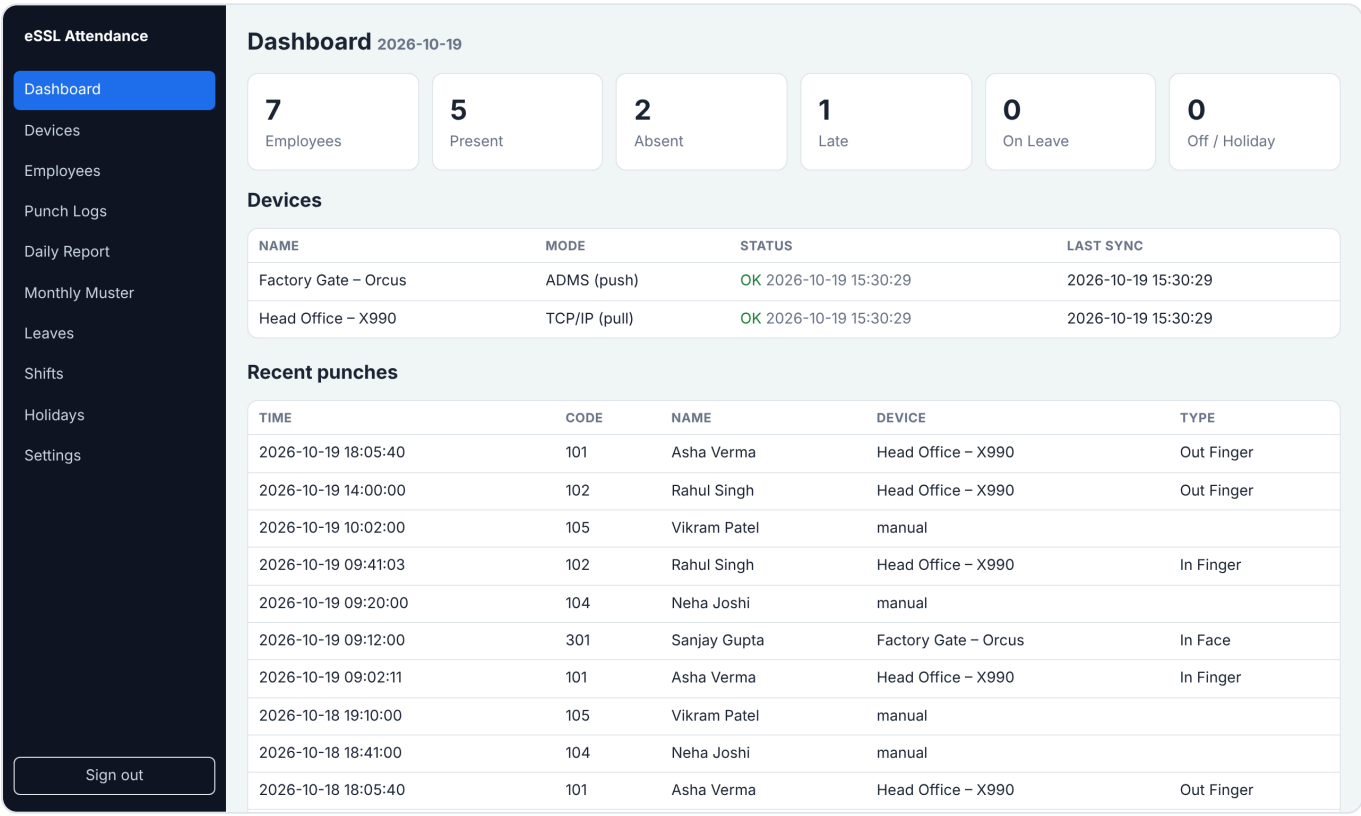
3.1 Sign in



Only the admin can sign in. The first admin account is created automatically the first time the app starts, with the username and password set by IT in the settings file.

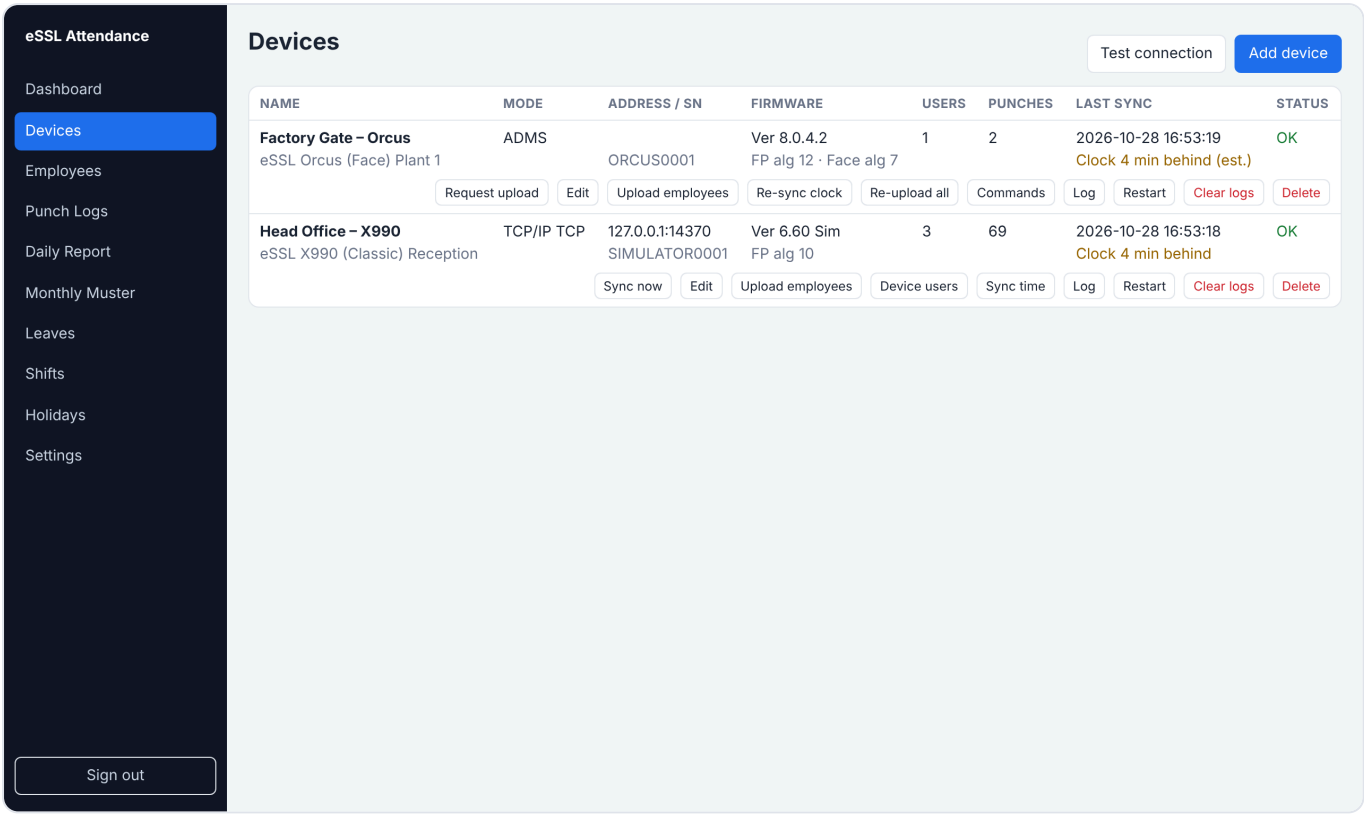
Sessions expire after 12 hours (adjustable). Passwords are stored encrypted (scrypt hash), never as plain text.

3.2 Dashboard



Today at a glance: how many employees are present, absent, late or on leave, whether each machine is connected, and the latest punches with the machine and method used (finger, face, card).

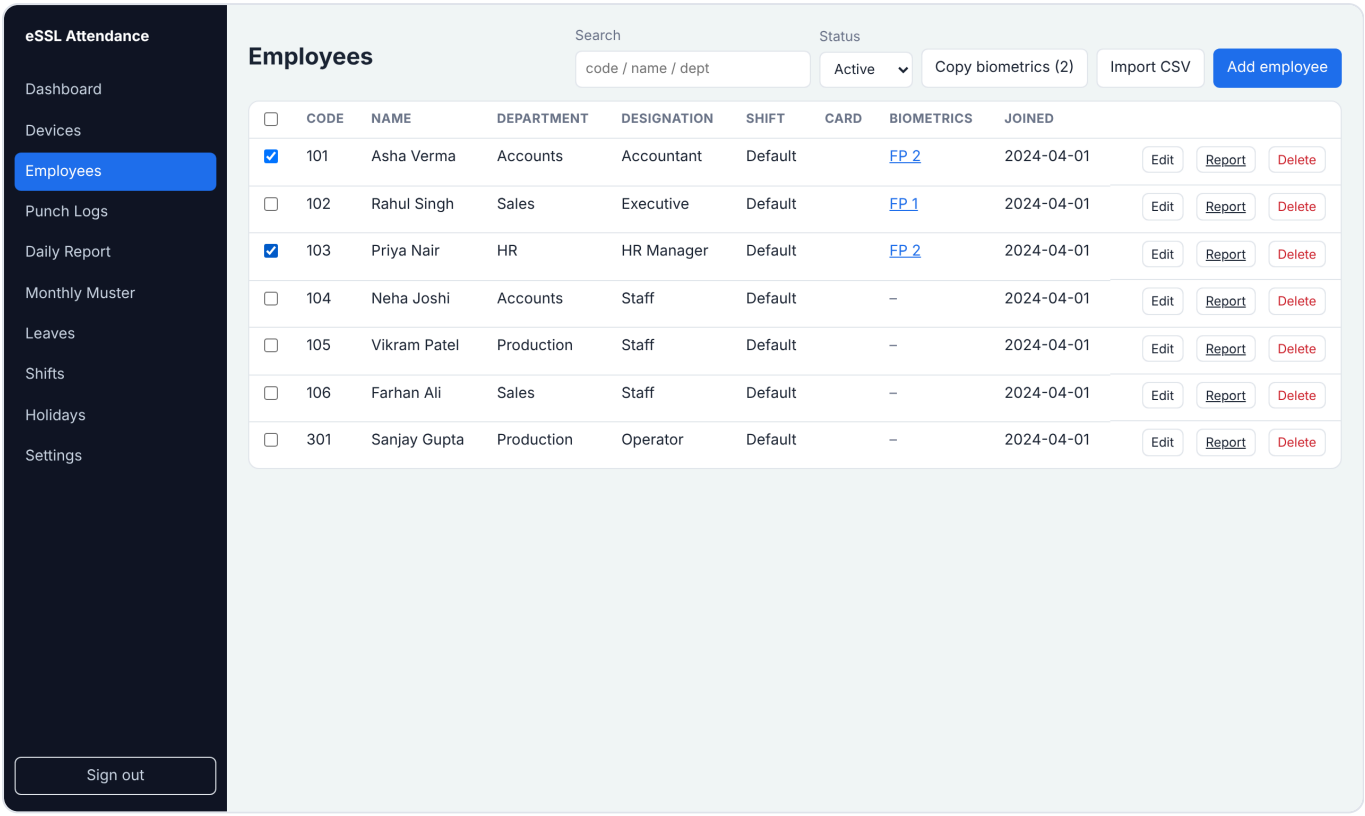
3.3 Devices



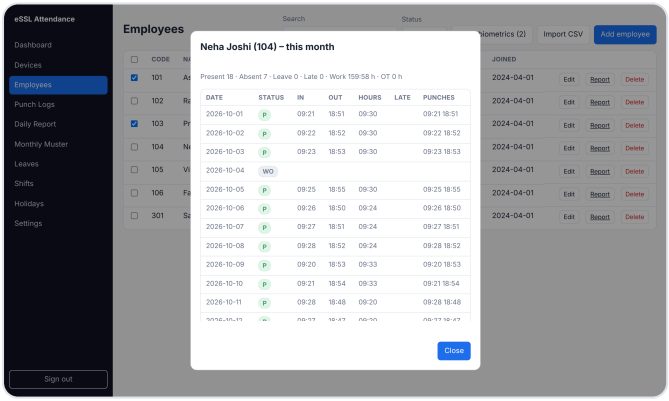
Each machine shows its mode, address or serial number, firmware, number of users and punches, last sync time, clock status (e.g. "Clock 4 min behind") and status. The buttons under each machine perform actions on it.

BUTTON	WHAT IT DOES
Sync now / Request upload	Downloads punches immediately (TCP/IP) or asks the machine to send any pending data (ADMS).
Upload employees	Sends employee ID, name and card number to the machine so their fingerprint or face can then be enrolled on it.
Device users	Lists the users stored on the machine (TCP/IP only).
Sync time / Re-sync clock	TCP/IP: sets the machine's clock and tells you how far off it was (also done automatically on every sync when it is off by more than a minute). ADMS: restarts the machine so it takes the server's time.
Re-upload all / Commands	ADMS only: ask the machine to resend its stored logs, or see and queue raw commands.
Log	History of syncs and errors for that machine.
Clear logs	Erases punches from the machine's memory, always after downloading them first.

3.4 Employees



Employee code (the same as the User ID on the machine), name, department, designation, shift, card and joining date. Search, filter active/inactive, add, edit, or import from CSV. Users already enrolled on a machine are imported automatically. The **Biometrics** column shows how many fingerprint (FP) and face templates the app holds for each person; tick the boxes on the left to choose people for **Copy biometrics** (next page).



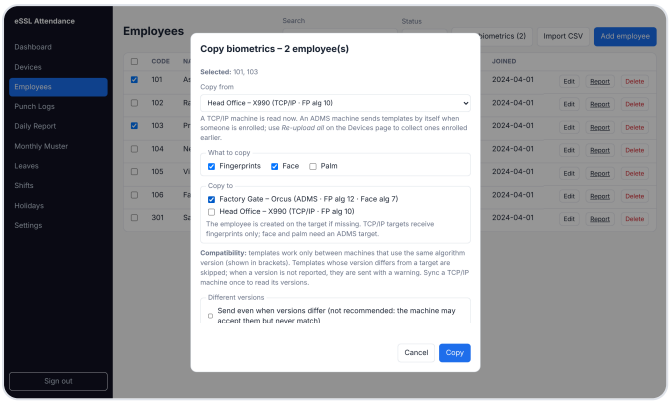
"Report" opens one employee's month: status, in/out, hours and every punch.

> **The employee code is the link**
>
> The code in the app must be the same as the User ID on the machine. Punches from an unknown User ID automatically create a placeholder employee ("Employee 105") that you can then rename and complete.

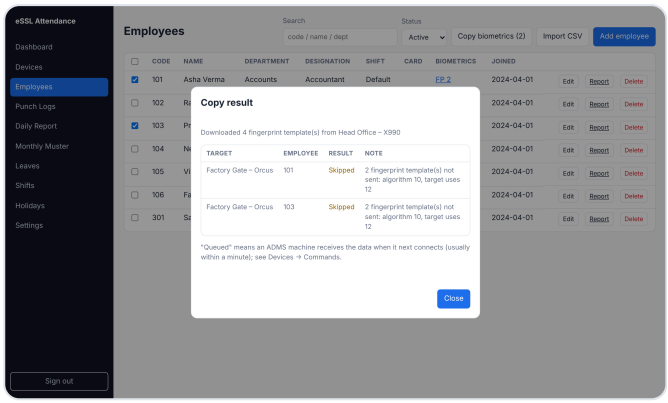
> **Deleting vs. inactive**
>
> Mark people who leave as *inactive* rather than deleting them, so old reports stay complete.

3.4.1 Copying biometrics of selected employees

Enrol a person's fingerprint or face once, then send it to the other machines from the app. Only the employees you tick are copied; everyone else on the target machines is left untouched.



Choose where to copy from (a machine, or the copies the app already holds), what to copy, and one or more target machines.



Result for each machine and employee: **Copied** (TCP/IP machine, done now), **Queued** (ADMS machine, delivered when it next connects) or **Skipped** with the reason.

1. **Employees**: tick the people to copy (the box in the header selects everyone in the list).
2. Click **Copy biometrics (N)**. Under **Copy from**, pick the machine where they were enrolled.
3. Tick **Fingerprints**, **Face** and/or **Palm**, and the target machines under **Copy to**. Click **Copy**.
4. Check the result table, then let one copied person verify on each target machine.

FROM ↓ TO →	TCP/IP MACHINE	ADMS MACHINE
TCP/IP machine	Fingerprints, copied immediately	Fingerprints, queued
ADMS machine	Fingerprints only (face/palm are skipped with a note)	Fingerprints, face and palm, queued

Same algorithm only, checked automatically

A template only works on a machine with the same matching algorithm (e.g. ZKFinger 10 on both). The app reads each machine's fingerprint and face algorithm versions (Devices page, under Firmware) and skips templates whose version differs from the target, saying why. Unknown versions are sent with a warning; test one person first.

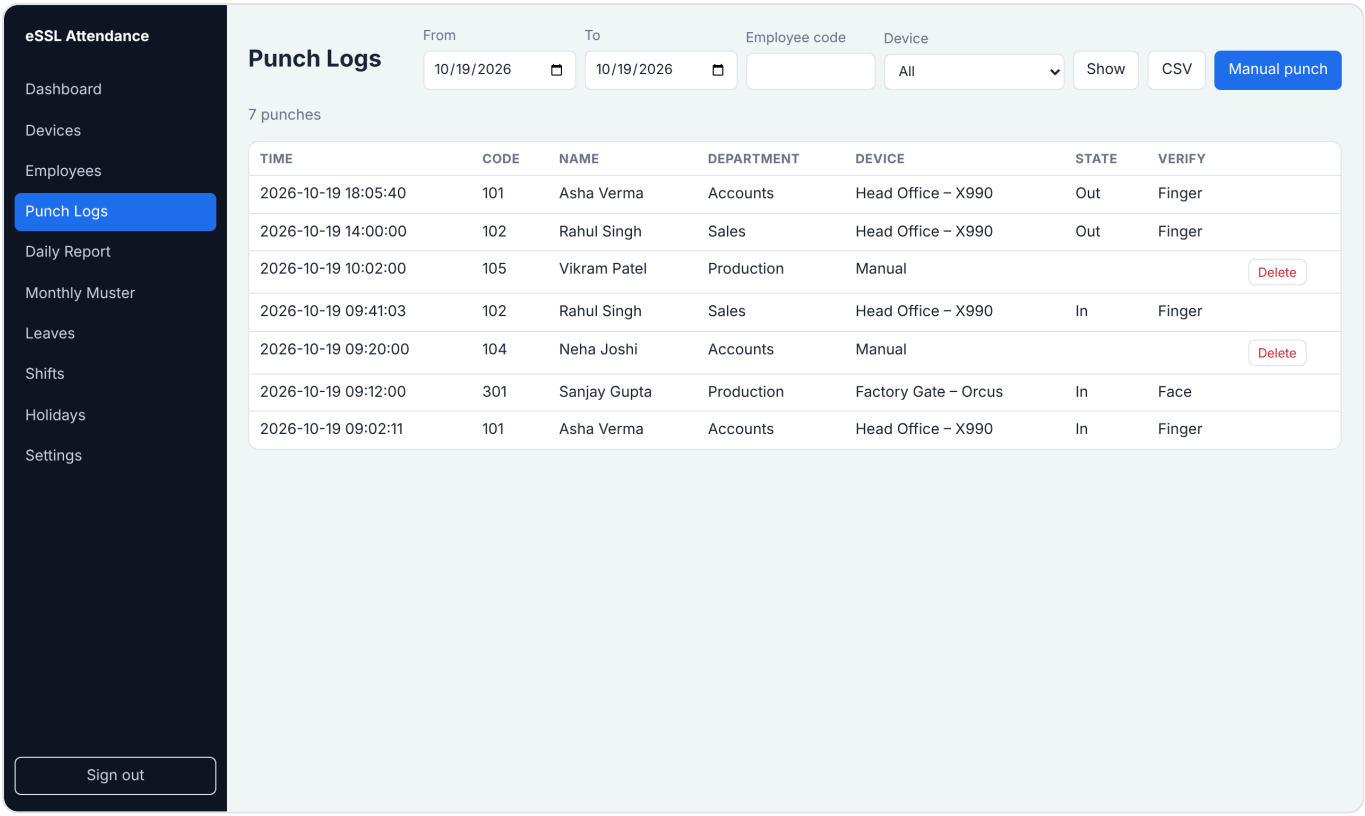
Where ADMS templates come from

ADMS machines send templates to the app by themselves when someone is enrolled. For people enrolled before the machine was connected, click **Re-upload all** on the Devices page once.

Sensitive data

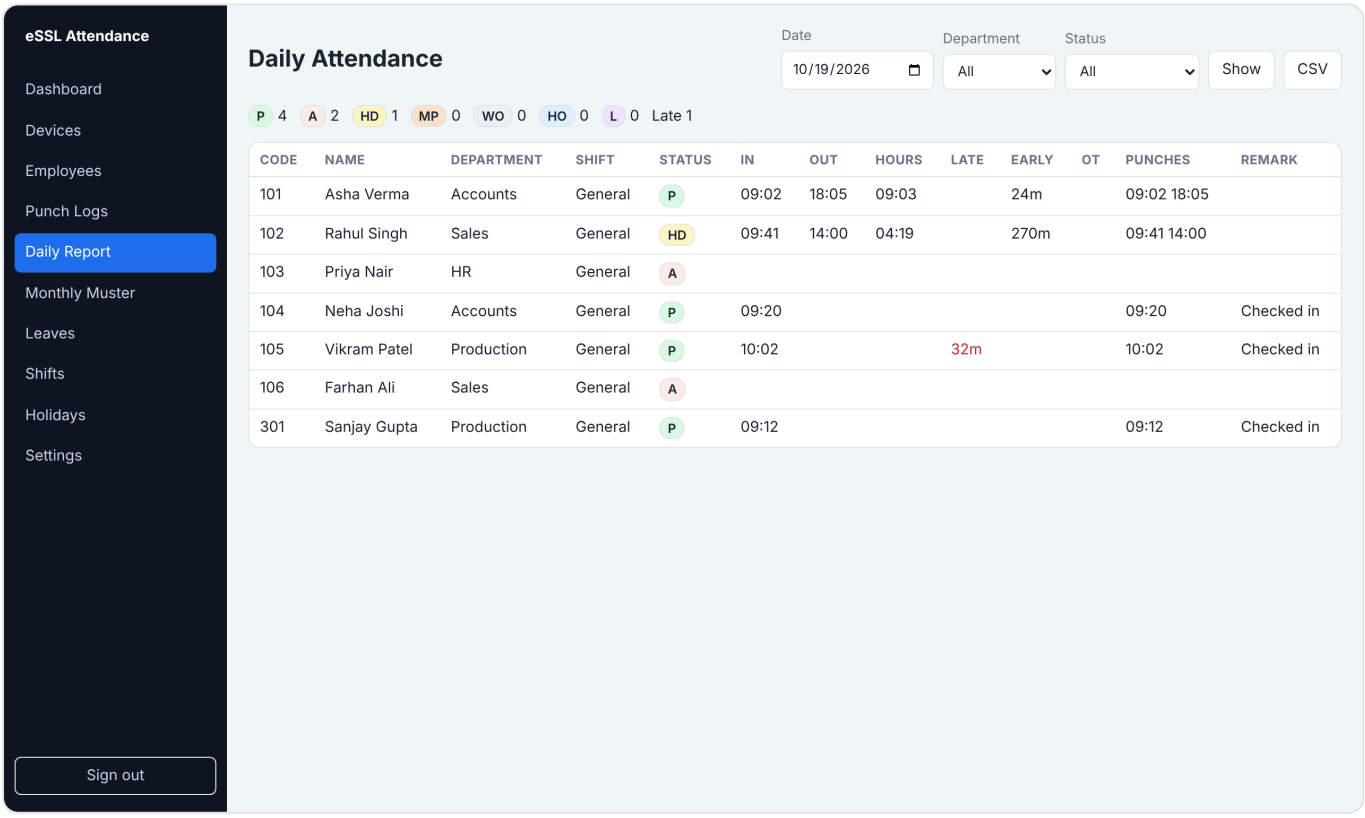
The app keeps a copy of each template in its database so it can be sent later. Click the count in the Biometrics column to see or delete one person's copies. Protect and back up the database file accordingly.

3.5 Punch Logs



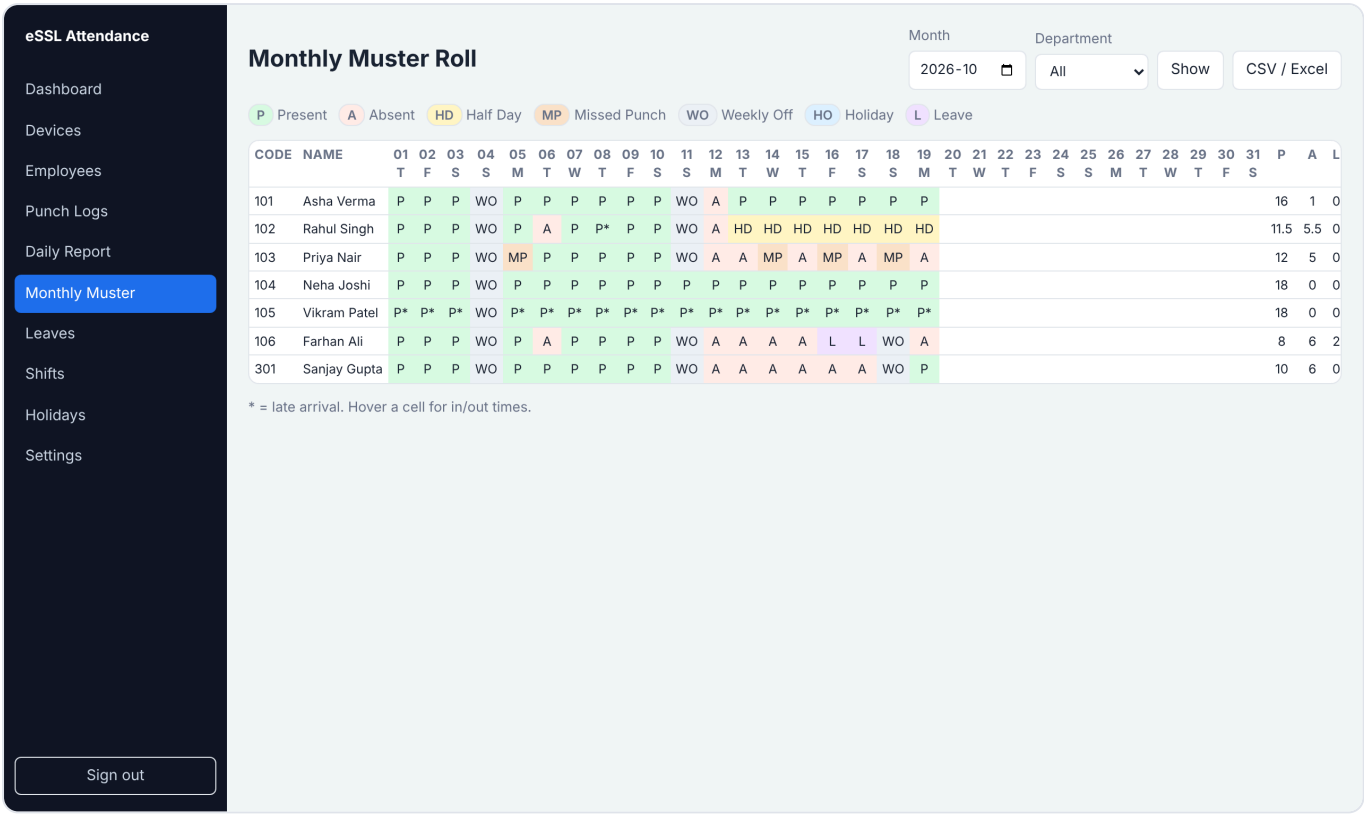
Every punch received, filterable by date range, employee and machine, with In/Out state and verification method. "Manual punch" adds a missing punch with a reason (for example, a forgotten check-out). Only manual punches can be deleted; machine punches cannot.

3.6 Daily Report



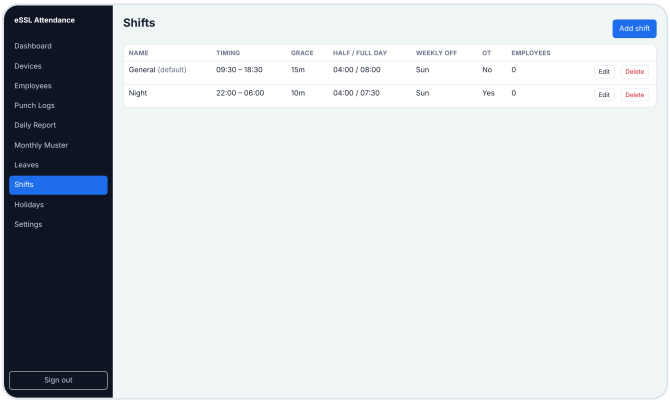
For any date: status, first In, last Out, hours, late minutes, early-exit minutes, overtime, all punches and a remark. Filter by department or status (for example "Late" or "Absent"). Export to CSV.

3.7 Monthly Muster Roll

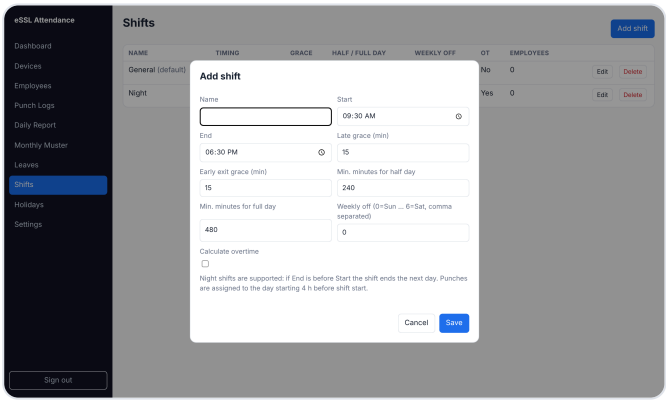


One row per employee, one coloured cell per day, plus totals at the right: Present, Absent, Leave, Weekly Off, Holiday, Late count, Payable days, Hours and Overtime. A * marks a late arrival. "CSV / Excel" downloads the whole sheet for payroll.

3.8 Shifts

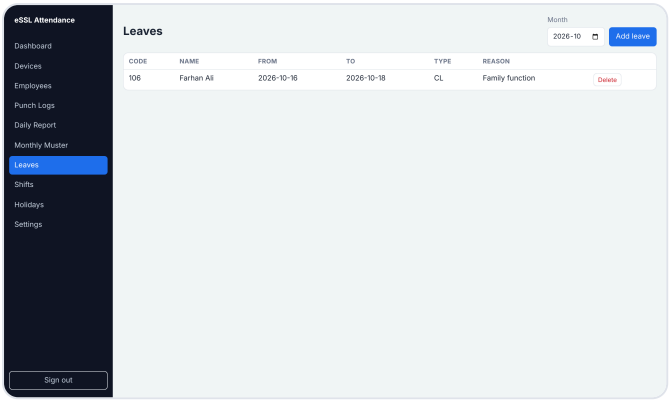


The first shift is the default for anyone without a shift. A night shift (22:00–06:00) is supported.

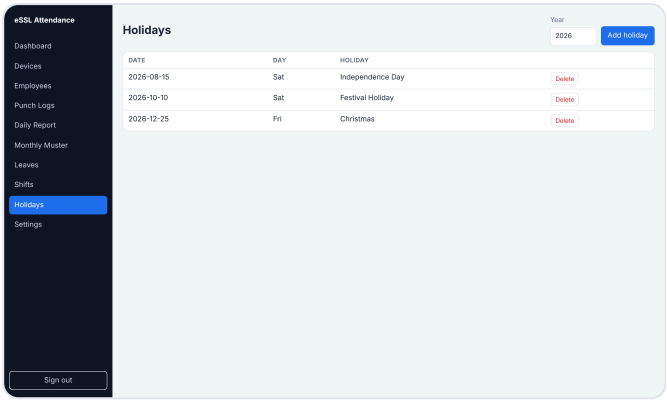


Each shift has start and end time, late grace, early-exit grace, minimum minutes for half and full day, weekly offs and an overtime switch.

3.9 Leaves and Holidays

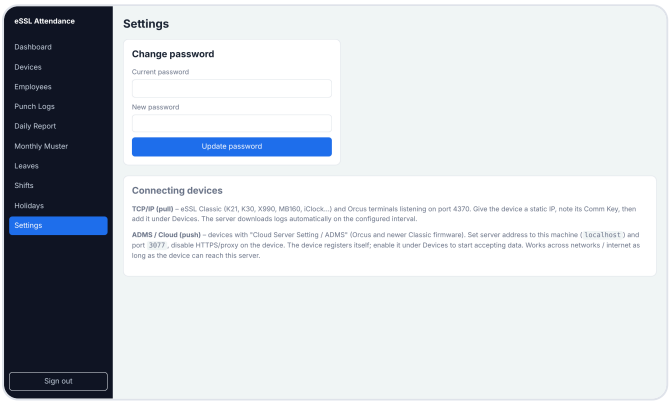


Leaves: CL, SL, EL, OD or leave without pay, for a date range, with a reason.

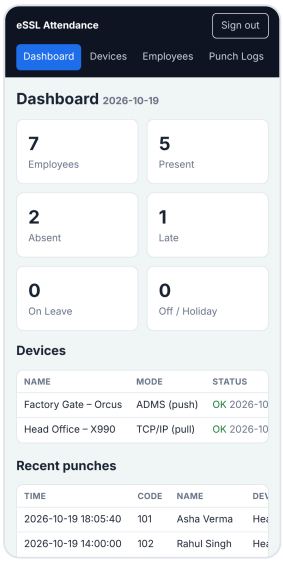


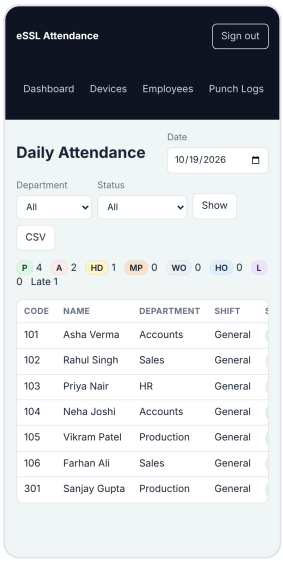
Holidays for the year. They show as HO in reports.

3.10 Settings and phone view



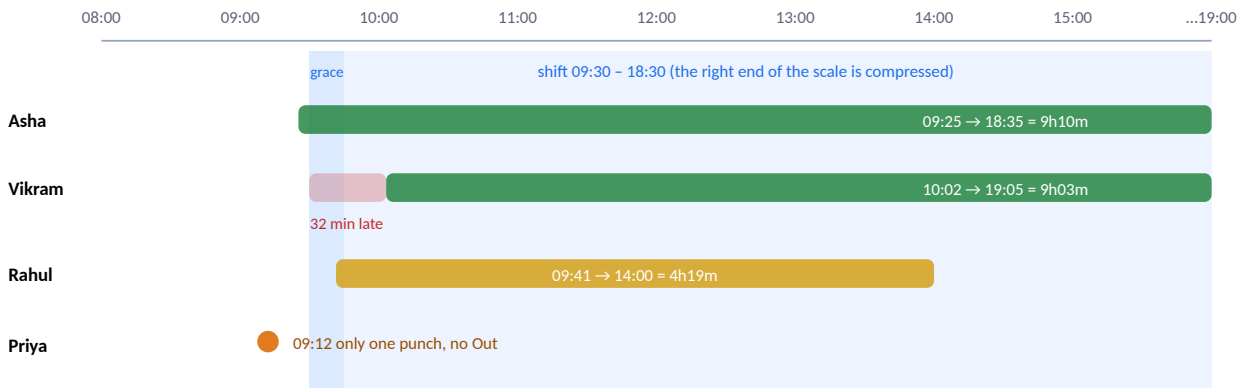
Change the admin password, plus built-in help on connecting machines.





4. How attendance is calculated

The app looks at all punches of an employee for a day, takes the **first punch as In** and the **last punch as Out**, and compares them with the employee's shift. Example: General shift 09:30–18:30, 15 minutes grace, half day = 4 hours, full day = 8 hours.



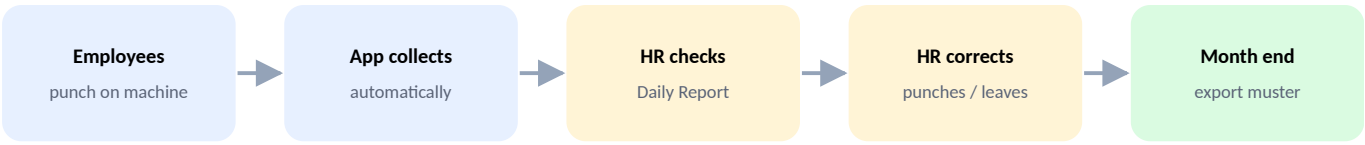
EMPLOYEE	WHAT HAPPENED	RESULT
Asha	Worked 9h10m, arrived before 09:45	P Present, on time
Vikram	Worked 9h03m but arrived at 10:02, after the grace period	P Present, marked late (32 min, shown as P*)
Rahul	Worked 4h19m: more than half a day, less than a full day	HD Half day, early exit 270 min
Priya	Only one punch on a past day	MP Missed punch (counted present; HR adds the missing punch)

Status codes

CODE	MEANING	WHEN IT IS GIVEN	COUNTS AS
P	Present	Worked at least the shift's full-day minutes	1 day
HD	Half day	Worked at least the half-day minutes but less than a full day	0.5 present, 0.5 absent
MP	Missed punch	Only one punch on a past day; today it shows as P until the day ends	1 day (to correct)
A	Absent	No punch on a working day, or worked less than half a day	0
WO	Weekly off	The shift's weekly-off day (Sunday by default) with no punches	1 paid day
HO	Holiday	A date on the Holidays list with no punches	1 paid day
L	Leave	An approved leave entry and no punches	1 paid day

Payable days = present days (half day = 0.5) + leave + weekly offs + holidays. **Overtime** (if switched on for the shift) = time worked beyond the shift length; all work done on a weekly off or holiday counts as overtime. Punches less than 1 minute apart count once. **Night shifts:** each attendance day starts 4 hours before the shift starts, so 22:00–06:00 work counts for the day it began, and a late check-out after midnight on a day shift still counts for the previous day.

5. Daily and monthly routine for HR



Every day (5 minutes)

1. Open **Dashboard**. Check that every machine shows **OK** and recently synced.
2. Open **Daily Report** for yesterday. Filter by **Missed Punch** and **Absent**.
3. For a forgotten punch: **Punch Logs → Manual punch**, enter the time and a reason.
4. For a leave: **Leaves → Add leave**. The day changes from A to L.

Every month

1. Make sure all holidays for the month are on the **Holidays** page.
2. Open **Monthly Muster**, choose the month, and review red (A) and orange (MP) cells.
3. Fix anything wrong (manual punches, leaves), then reload.
4. Click **CSV / Excel** and send the file to payroll.
5. IT: keep a copy of the database file as a backup.

When someone joins

1. **Employees → Add employee**: choose a code (e.g. 107), name, department, shift and joining date.
2. **Devices → Upload employees**, enter `107`. The user appears on the machine (for ADMS machines, within a few seconds).
3. On one machine, enrol the fingerprint or face for user 107 (Menu → User Mgt). Then, in the app, tick 107 on **Employees** and use **Copy biometrics** to send it to the other machines.

When someone leaves

Edit the employee and untick **Active**. Optionally remove them from the machine (on TCP/IP machines, delete the user on the machine itself or through the API).

6. Setting up the machines

Method A: TCP/IP

1. On the machine: **Menu → Comm. → Ethernet**. Set a fixed IP (e.g. 192.168.1.201), subnet and gateway.
2. **Comm. → PC Connection**: note the **Comm Key** (default 0) and port (4370).
3. In the app: **Devices → Test connection** with that IP. You should see the serial number and firmware.
4. **Add device**, mode *TCP/IP*, same IP, port and comm key. Click **Sync now**.

Choose protocol UDP only for very old firmware that refuses TCP.

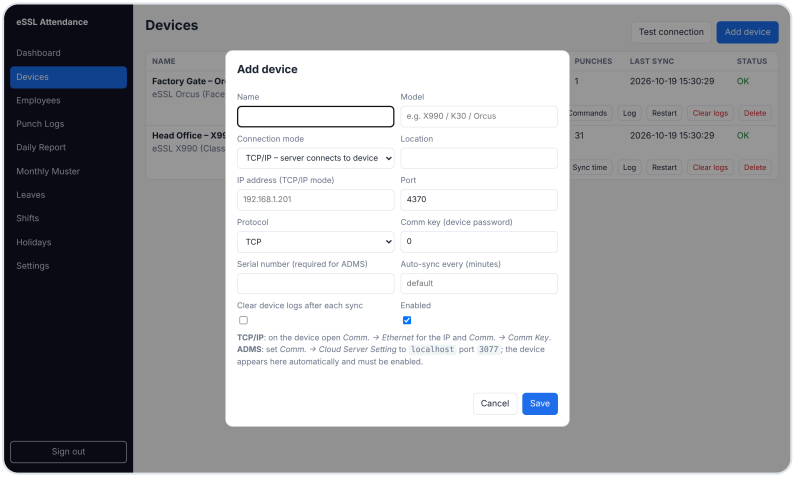
Method B: ADMS / Cloud Server

1. On the machine: **Menu → Comm. → Cloud Server Setting** (sometimes called ADMS).
2. Server address: your app's domain or IP (e.g. `attendance.company.com`). Port: `80` (or the app's port). HTTPS: **off**. Proxy: **off**.
3. Wait a minute. In the app, **Devices** shows "ADMS <serial number>".
4. Click **Edit**, give it a name, tick **Enabled**, Save. Punches start arriving.

Until it is enabled the app does not accept the machine's data, so the machine keeps it and resends it after approval. Nothing is lost.

Time zone and clock drift

ADMS machines take the server's time and time zone (`ADMS_TIMEZONE` , `330` = India) only when they connect, so their clock can slowly fall behind. The Devices page shows the estimated error; **Re-sync clock** fixes it, and **Edit → Restart daily at** (e.g. 03:30) keeps it right automatically. TCP/IP machines are corrected on every sync.



The app's "Add device" form (TCP/IP machines). A new ADMS machine adds itself, and you only need to tick **Enabled**.

Menu names differ slightly between models

LOOK FOR	ALSO CALLED
Comm. → Ethernet	Network, TCP/IP, Comm Settings
Comm Key	Comm Password, Device Password (in PC Connection)
Cloud Server Setting	ADMS, Server Settings, Push, iClock Server
User Mgt	User Management, Enroll User

7. Putting it online (GoDaddy)

The app is a Node.js web application with its data in one file (SQLite), so it does not need a separate database server. Which GoDaddy product you have decides how it is installed.

GODADDY PRODUCT	WORKS?	NOTES
VPS or dedicated server	Yes	Recommended. Full control, always running, no limits on the database library.
cPanel hosting with "Setup Node.js App"	Probably	Works if your cPanel shows this option and if *Run NPM Install* succeeds (it must build the database library). Not yet tried on a real GoDaddy account.
cPanel without Node.js, Website Builder, WordPress hosting	No	These cannot run Node.js apps. Use a VPS instead.

cPanel steps (short version)

1. **Setup Node.js App → Create**: Node 20 or newer, startup file `app.js`, URL at the root of a subdomain (e.g. attendance.company.com).
2. Upload all project files except `node_modules` and `data`.
3. Add environment variables: `ADMIN_PASSWORD`, `ADMS_TIMEZONE=330`, `DB_PATH` (a folder outside public_html).
4. **Run NPM Install**, then **Restart**.
5. Do **not** turn on "Force HTTPS" for this subdomain: machines send data over plain HTTP.

VPS steps (short version)

1. Install Node.js 22, nginx and build tools.
2. Copy the code, run `npm ci --omit=dev`, create `.env`.
3. Start with `pm2 start app.js --name essl` so it restarts automatically.
4. Point the subdomain to the VPS; nginx forwards ports 80/443 to the app.
5. Keep plain HTTP working for `/iclock/`; add HTTPS for the web screens.

Settings file (.env) at a glance (exact install commands are in the README)

SETTING	WHAT IT CONTROLS	DEFAULT
`ADMIN_USER`, `ADMIN_PASSWORD`	First admin login (used only on the very first start)	admin / admin123
`PORT`	Web port of the app	3000
`DB_PATH`	Where the database file is kept	data/attendance.db
`POLL_MINUTES` / `AUTO_TIME_SYNC` / `TIME_DRIFT_SECONDS`	How often TCP/IP machines are synced / correct their clocks / allowed error before correcting	5 / true / 60
`ADMS_AUTO_APPROVE`	Accept new ADMS machines without approval	false
`ADMS_TIMEZONE`	Time zone sent to ADMS machines	330 (IST)

8. What is done

Everything below is built, in the code repository, and covered by tests or checked by hand in the browser.

AREA	DELIVERED	STATUS
Machine link: TCP/IP	Connect on port 4370 over TCP or UDP, comm-key login, read serial/firmware/counts, download all punches (old and new record formats), download users, add/update/delete users, set clock, restart, clear logs after saving	Done
Machine link: ADMS	Handshake, live punch upload, user upload from machine, device info, command queue (add/delete user, restart, clear logs, resend), approval of new machines	Done
Automatic sync	Background schedule per machine, one operation at a time per machine, clock checked on every sync and corrected when off by more than a minute, duplicate punches ignored	Done
Employees	Add/edit/delete, departments, designation, card, shift, joining date, active flag, CSV import, auto-import from machines	Done
Rules	Multiple shifts, grace, early exit, half/full day, weekly offs, overtime, night shifts, holidays, leaves (CL/SL/EL/OD/LWP), manual punches with reason	Done
Reports	Dashboard, daily report, monthly muster with payable days, per-employee report, punch log; CSV export (opens in Excel)	Done
Security	Admin login, encrypted password, secure session cookie, password change, protection against harmful text in exported spreadsheets	Done
Screens	10 screens, phone-friendly, light and dark mode	Done
Hosting	Runs on Windows, Linux, macOS, a GoDaddy VPS, and cPanel Node.js hosting (startup file `app.js`, Node 20+)	Done
Copy biometrics	Copy fingerprint, face and palm templates of selected employees from one machine (or the stored copies) to several machines; TCP/IP and ADMS; reads each machine's algorithm versions and skips incompatible templates; result per employee; view/delete stored copies	Done
Clock drift	Shows each machine's clock error on the Devices page (read from TCP/IP machines, estimated from live punches for ADMS); one-click correction; optional daily restart for ADMS machines; answers the ADMS server-time request	Done
Testing aids	Simulators for a TCP/IP machine and an ADMS machine; 18 automated tests	Done
Documentation	README with install, machine setup, GoDaddy deployment and API; this guide	Done

Fixed while preparing this guide

Hours and late minutes could show long decimals (e.g. "24.333m"), the device action buttons overflowed the table, and information pop-ups showed both Cancel and Close. All three are fixed in the latest version.

9. What still needs to be done

9.1 Before go-live (must do)

#	TASK	WHY	WHO
1	**Test with a real Classic machine** (sync, users, upload, clock)	The app has only been tested against simulators built from protocol documentation. Firmware versions differ.	IT + developer
2	**Test with a real Orcus machine** in ADMS mode	Same reason. Confirm punches arrive, the face verification code is shown, and the clock and time zone are right.	IT + developer
2a	**Test copying biometrics** between your actual models (Classic → Classic, Orcus → Orcus, Classic → Orcus)	The template format was built from the protocol documentation and checked with simulators only. Confirm a copied person can punch on each target.	IT + developer
2b	**Check the Orcus clock** after a restart and after a day	Confirms it takes the server time on connect (time and time zone) and how fast it drifts, to choose whether a daily restart is needed.	IT
3	**Choose hosting** (VPS recommended) and install	cPanel Node.js hosting may not allow the database library to build.	IT
4	**Set a strong admin password** and change the default	The default password is public in the documentation.	IT
5	**Enable HTTPS** for the web screens	So logins are encrypted. Keep plain HTTP open only for machine uploads (`/iclock/`).	IT
6	**Schedule daily backups** of the database file	All attendance history lives in one file.	IT
7	**Enter master data**: shifts, weekly offs, holiday list, employees with the same codes as on the machines	Reports are only as good as the setup.	HR
8	**Run in parallel for 1–2 weeks** and compare with the current method	To catch any rule that does not match company policy before payroll depends on it.	HR

9.2 Recommended improvements (next phase)

FEATURE	BENEFIT	PRIORITY	EFFORT
Multiple users and roles (HR, manager, viewer)	Today there is one admin login. Department managers could see only their team.	High	Medium
Audit log of changes	Records who added a manual punch or changed a leave. Important for disputes and audits.	High	Small
Lock a month after payroll	Stops anyone changing attendance that has already been paid.	High	Small
Automatic database backup inside the app	Daily copy without needing IT to set it up.	High	Small

FEATURE	BENEFIT	PRIORITY	EFFORT
Leave balances and approval	Yearly quota per leave type, balance shown, request and approve.	Medium	Medium
Excel (.xlsx) and PDF reports	Formatted, printable muster and daily reports with company name.	Medium	Small
Email / WhatsApp alerts	Daily absent list to HR, alert when a machine stops sending data.	Medium	Medium
Payroll export format	Export in the exact layout your payroll software or CA needs.	Medium	Small
Rotating shifts and shift roster	For factories where people change shift weekly.	Medium	Medium
Branches / locations	Separate holidays and reports per site.	Medium	Medium
Clock setting for ADMS machines	A reliable command to set the clock from the app (today ADMS machines take time at connection).	Medium	Small*
Employee self-service	Staff see their own attendance and apply for leave on their phone.	Low	Large
Comp-off, OT approval, permission hours	Finer HR policies.	Low	Medium

Effort: Small = about 1–2 days, Medium = about 3–7 days, Large = more than a week. * Needs testing on real machines because firmware behaviour varies.

10. Troubleshooting and FAQ

PROBLEM	WHAT TO CHECK
Test connection: "timed out" or "ECONNREFUSED"	The PC running the app must reach the machine: ping its IP; same network; port 4370 not blocked by a firewall; correct IP on the machine.
"Device rejected the communication key"	The Comm Key in the app must match Comm. → PC Connection on the machine.
ADMS machine never appears	Server address and port on the machine; HTTPS and proxy off; the machine has internet; "Force HTTPS" is off on the hosting; the app is at the root of the domain.
ADMS machine appears but no punches	It is not enabled yet: Devices → Edit → tick Enabled.
Machine clock runs slow (e.g. Orcus)	TCP/IP: corrected on every sync. ADMS: click *Re-sync clock*, and set *Restart daily at* in the machine's settings. If it is wrong by a fixed amount (e.g. 5 h 30 min) right after connecting, check `ADMS_TIMEZONE`.
"Employee 105" appears in lists	A user punched who is not in Employees yet. Edit it and enter the real name and details.
Someone shows Absent but was present	Check Punch Logs for that date. If the punch is missing, add a manual punch with a reason.
Copy says "not sent: algorithm 10, target uses 12", or a copied person cannot punch	The machines use different fingerprint/face engines (compare the versions on the Devices page). Enrol the person on that machine directly.
Copy result says "No stored biometrics"	Pick the enrolling machine under *Copy from*. For an ADMS machine, click *Re-upload all* on the Devices page first and wait a minute.
Night-shift worker shows two half days	Assign them the Night shift on their employee record.
Machine memory is full	Devices → Clear logs (it downloads first), or tick "Clear device logs after each sync".
Forgot admin password	IT can delete the users entry in the database, or start with a new database. Ask the developer for a reset script (a small future task).

Frequently asked

Does the PC need to stay on? For TCP/IP, yes, while you want punches downloaded (the machine keeps them meanwhile). For ADMS on GoDaddy, no office PC is needed.

How many employees and machines? No fixed limit; several hundred employees and years of punches are fine.

Can I still use eSSL eTimeTrackLite at the same time? Usually, for TCP/IP machines (untested): the next sync retries if the machine is busy, and neither program may clear the machine's logs. An ADMS machine sends data to only one server.

Can data move between Classic and Orcus machines? User records (ID, name, card) yes, with *Upload employees*. Fingerprints only if the Orcus has a fingerprint sensor and both report the same fingerprint algorithm version; the copy dialog checks this. Faces no: Classic machines have no compatible face templates, so people enrol their face once on an Orcus and the app can copy it to other Orcus machines with the same face engine.

Is the data safe? It stays on your server. Logins are protected, but back up the database file regularly (section 9).

11. Glossary

TERM	MEANING
Punch	One fingerprint, face or card scan on the machine, with date and time.
Employee code / User ID / PIN	The number that identifies a person on the machine. It must match the code in the app.
TCP/IP (pull) mode	The app connects to the machine over the office network on port 4370 and downloads data.
ADMS / Cloud Server (push) mode	The machine connects to the app over the internet and sends data itself.
Comm Key	A numeric password on the machine that protects TCP/IP connections.
Serial number (SN)	Unique ID of the machine, used to recognise ADMS machines.
Sync	Downloading new punches and users from a machine.
Grace time	Minutes after shift start before someone is marked late.
Muster roll	Monthly sheet with one row per employee and one column per day.
Payable days	Days the employee is paid for: present + paid leave + weekly offs + holidays.
Manual punch	A punch entered by HR to correct a forgotten or failed scan.
CSV	A simple spreadsheet file that opens in Excel.
Node.js	The software platform the app runs on.
SQLite	The database: a single file holding all the app's data.
VPS	Virtual private server: a rented server where you control the software.
Template	The machine's coded form of a fingerprint or face (not a photo). It is what gets copied between machines.
Simulator	A program that imitates a machine, used for testing without hardware.

> **Where to find more**
>
> Technical details (all settings, the REST API, project layout and exact installation commands) are in `README.md` in the code repository. The automated tests run with `npm test`.